Practical 1

Aim : Study and perform practical on Cryptography algorithm for securing data.

To study cryptographic techniques used to secure data during storage and transmission, and implement basic encryption and decryption algorithms.

Theory:

Cryptography is the process of converting readable data (plaintext) into an unreadable form (ciphertext) so that only authorized users can access it. It helps ensure confidentiality, integrity, and secure communication.

- **Types of Cryptography:**
  - **Symmetric Key Cryptography**: Uses the same key for encryption and decryption. Examples: Caesar Cipher, Vigenère Cipher, AES, DES.
  - **Asymmetric Key Cryptography**: Uses a public key for encryption and a private key for decryption. Examples: RSA, Diffie-Hellman.

Code:

```
import hashlib
def encrypt(text, shift):
    result = ""
    for ch in text:
        if ch.isalpha():
            base = ord('A') if ch.isupper() else ord('a')
            result += chr(
                (ord(ch) - base + shift) % 26 + base
            )
        else:
            result += ch
    return result

def decrypt(text, shift):
    return encrypt(text, -shift)
message = "Hello Blockchain"
key = 3
encrypted = encrypt(message, key)
decrypted = decrypt(encrypted, key)
```

```
digest = hashlib.sha256(
    message.encode("utf-8")
).hexdigest()
print("Original:", message)
print("Encrypted:", encrypted)
print("Decrypted:", decrypted)
print("SHA-256:", digest)
```

Output:

```
PS D:\ADIT_DESKTOP\SEM - 7th\BCHAIN> python practical_1.py
Original: Hello Blockchain
Encrypted: Khoor Eorfnfkdlq
Decrypted: Hello Blockchain
SHA-256: 7cf88f2ee398c0b7c0e760a1dccaf3571e0baccf310f11fe3bdfd0b09675ea75
```